# BIODIESEL PRODUCTION PLANT: PROCESS SIMULATION REPORT

This report documents the steady-state simulation of a continuous biodiesel plant that produces fatty acid methyl esters (FAME) by alkali-catalysed transesterification of a triglyceride-rich oil with methanol. The basis is the process flow diagram and the stream material table supplied with this request. The report covers the process description, feed specifications, operating conditions, simulation results, product quality, mass and energy balances, observations on the design, and the assumptions and limitations that apply to the results.

## 1. EXECUTIVE SUMMARY

The plant processes 1,050.0 kg/h of oil with 127.1 kg/h of fresh methanol and produces 1,051.8 kg/h of FAME, 114.2 kg/h of glycerol and 133.4 kg/h of a water-methanol stream. The key results are listed below.

- **Product quality.** The FAME stream is 99.7 wt% ester, with monoglyceride as the main impurity at 0.27 wt%. The predicted values meet the EN 14214 limits that could be checked.
- **Reaction performance.** One-pass conversion of glyceride ester groups in the reactor is 96.4%. Unreacted glycerides are recycled, and 99.7% of the fresh ester groups leave the plant as methyl esters.
- **Methanol recovery.** The methanol column recovers 94.0% of the methanol leaving the reactor, which is recycled to the reactor. The total methanol-to-oil molar ratio at the reactor is about 6.0:1, of which 3.1:1 is fresh make-up.
- **Mass balance.** Every unit and the overall plant close to within rounding (overall error below 0.0001%).
- **Energy.** Net external heating is about 125 kW, dominated by the ESTCOL reboiler (68.9 kW). Net cooling is about 14 kW. The recycled oil cooler releases 9.8 kW that could be used to preheat fresh oil.

## 2. PROCESS DESCRIPTION

The flowsheet has five functional sections: catalyst and methanol preparation, transesterification, methanol recovery, ester washing and purification, and catalyst neutralisation with glycerol recovery. Figure 1 shows the flowsheet as supplied.

### *2.1 Methoxide preparation*

Fresh methanol and a 20 wt% sodium hydroxide solution are mixed in MIXER1 at 25 °C and 1.0 bar. The mixture (MIX1) warms to 33.4 °C, which reflects the heat of mixing and methoxide formation. PUMP1 raises the pressure to 4.0 bar, and MIXER2 combines the stream with recycled methanol (MEOHREC) to give the reactor feed MIX3 at 32.2 °C.

### *2.2 Oil feed conditioning*

Fresh oil enters at 25 °C and 1.0 bar and is mixed in B3 with the recycled unreacted oil (OILREC2, 41.4 kg/h). PUMP2 raises the mixture from 0.2 bar to 4.0 bar, and HX1 heats it from 25.4 °C to 60.0 °C before it enters the reactor.

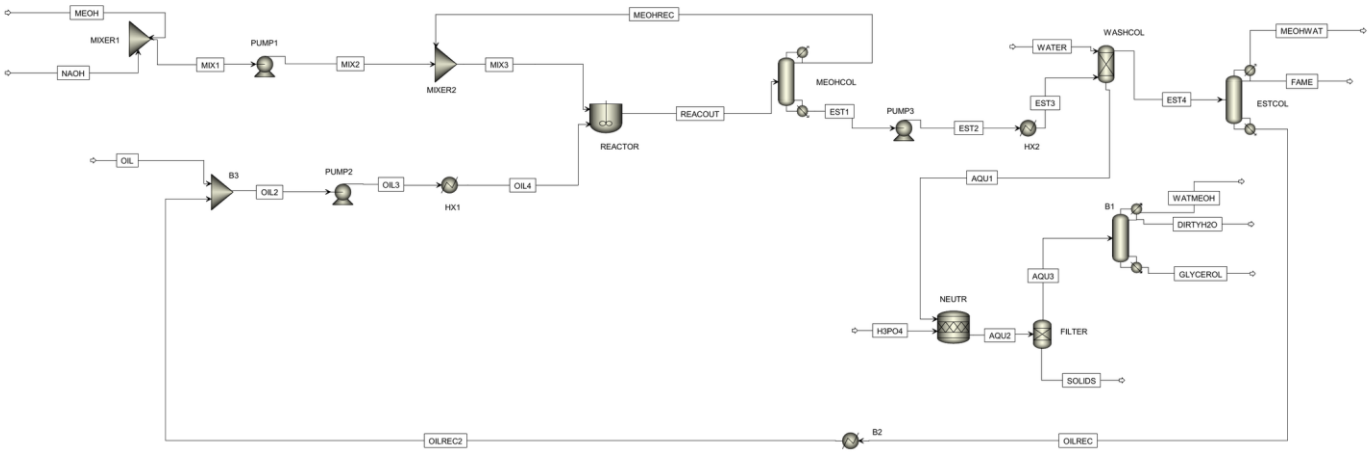



*Figure 1. Process flow diagram of the biodiesel plant*

### *2.3 Transesterification*

The reactor (REACTOR) is a stirred tank operated at 60.0 °C and 4.0 bar. Triglycerides react with methanol in the presence of the NaOH catalyst to form methyl esters and glycerol. The effluent REACOUT (1,391.3 kg/h) contains the ester product, unreacted methanol, glycerol, unreacted glycerides, catalyst and water.

### *2.4 Methanol recovery*

MEOHCOL is a vacuum distillation column operated at 0.2 bar with a condenser and reboiler. Methanol is recovered overhead at 98 wt% purity (MEOHREC) and returned to MIXER2. The bottoms (EST1, 1,268.5 kg/h at 60.0 °C) carry the esters, glycerol and catalyst to the washing section. PUMP3 raises this stream from 0.2 bar to 1.1 bar, and HX2 provides final temperature trimming.

### *2.5 Water washing*

WASHCOL is a counter-current water-wash column. Fresh water (50.0 kg/h at 25 °C) removes the glycerol, the NaOH and most of the residual methanol from the ester phase. The aqueous stream (AQU1) leaves the bottom at 60.9 °C, and the washed ester stream (EST4) leaves the top at 57.8 °C.

### *2.6 FAME purification*

ESTCOL is a vacuum column operated at 0.1 bar at the top. The FAME product is drawn at 148.3 °C. Light components, mainly water, leave overhead as MEOHWAT (4.9 kg/h). Residual glycerides and heavy esters leave as bottoms (OILREC, 41.4 kg/h) at 338.7 °C. These are cooled in B2 to 25 °C and recycled to B3.

### *2.7 Catalyst neutralisation and glycerol recovery*

The aqueous stream AQU1 is neutralised in NEUTR at 50 °C with 20 wt% phosphoric acid (H3PO4), which converts the NaOH to sodium phosphate (Na3PO4). FILTER removes the solid Na3PO4 (SOLIDS, 13.7 kg/h), and the clear aqueous stream AQU3 goes to column B1. B1 operates at 0.4 to 0.5 bar. It separates a water-methanol overhead (WATMEOH, 70.9 °C) from glycerol bottoms (GLYCEROL, 263.5 °C), which is the product.

## 3. EQUIPMENT AND OPERATING CONDITIONS

Table 1 lists the units in the flowsheet with their function, and Table 2 gives the operating conditions of the streams leaving each unit.

***Table 1: Equipment inventory***

| Unit | Type | Function |
|---|---|---|
| MIXER1 | Mixer | Forms methoxide feed from methanol and NaOH solution |
| PUMP1 | Pump | Raises methoxide feed from 1.0 to 4.0 bar |
| MIXER2 | Mixer | Combines methoxide with recycled methanol |
| B3 | Mixer | Combines fresh oil with recycled oil |
| PUMP2 | Pump | Raises oil feed from 0.2 to 4.0 bar |
| HX1 | Heat exchanger | Heats oil from 25.4 to 60.0 °C |
| REACTOR | Stirred tank | Transesterification at 60.0 °C and 4.0 bar |
| MEOHCOL | Vacuum column | Recovers methanol for recycle; bottoms to washing |
| PUMP3 | Pump | Raises ester stream from 0.2 to 1.1 bar |
| HX2 | Heat exchanger | Trim cooling before washing |
| WASHCOL | Water-wash column | Removes glycerol, NaOH and methanol to aqueous stream |
| ESTCOL | Vacuum column | Separates FAME side draw, overhead vapour and residual glycerides |
| B2 | Heat exchanger | Cools recycled glyceride stream to 25 °C |
| NEUTR | Neutraliser | Neutralises NaOH with H3PO4 |
| FILTER | Filter | Removes Na3PO4 solids |
| B1 | Distillation column | Separates water-methanol overhead from glycerol bottoms |

***Table 2: Operating conditions by unit***

| Unit | Outlet stream | T (°C) | P (bar) | Notes |
|---|---|---|---|---|
| MIXER1 | MIX1 | 33.4 | 1.0 | Heat of mixing |
| PUMP1 | MIX2 | 33.7 | 4.0 | Pump outlet |
| MIXER2 | MIX3 | 32.2 | 0.2 | Outlet at lowest inlet pressure |
| B3 | OIL2 | 24.9 | 0.2 | Oil and recycle mix |
| PUMP2 | OIL3 | 25.4 | 4.0 | Pump outlet |
| HX1 | OIL4 | 60.0 | 4.0 | Reactor feed heater |
| REACTOR | REACOUT | 60.0 | 4.0 | Transesterification |
| MEOHCOL | MEOHREC / EST1 | 28.8 / 60.0 | 0.2 | Top / bottom |
| PUMP3 | EST2 | 60.2 | 1.1 | Pump outlet |
| HX2 | EST3 | 60.0 | 1.1 | Trim cooler |
| WASHCOL | EST4 / AQU1 | 57.8 / 60.9 | 1.0 | Top / bottom |
| ESTCOL | FAME / MEOHWAT / OILREC | 148.3 / 148.3 / 338.7 | 0.1 / 0.1 / 0.2 | Side draw / top / bottom |
| B2 | OILREC2 | 25.0 | 0.2 | Recycle cooler |
| NEUTR | AQU2 | 50.0 | 1.1 | Neutralisation |
| FILTER | AQU3 / SOLIDS | 50.0 | 1.1 | Filtrate / solids |
| B1 | WATMEOH / GLYCEROL | 70.9 / 263.5 | 0.4 / 0.5 | Top / bottom |

# 4. FEED SPECIFICATIONS

Table 3 gives the five fresh feeds. The oil is the largest feed, at 1,050.0 kg/h, and has a mean molecular weight of 822 kg/kmol. The NaOH is supplied as a 20 wt% aqueous solution, and the phosphoric acid as a 20 wt% solution.

***Table 3: Fresh feed streams***

| Stream | T (°C) | P (bar) | Flow (kg/h) | Flow (kmol/h) | Composition |
|---|---|---|---|---|---|
| MEOH | 25.0 | 1.0 | 127.12 | 3.967 | 100 wt% methanol |
| NAOH | 25.0 | 1.0 | 50.00 | 2.470 | 20 wt% NaOH (10.00 kg/h NaOH) in water |
| OIL | 25.0 | 1.0 | 1,050.00 | 1.277 | Triglyceride-rich oil (92.5 mol% TG, 7.5 mol% DG) |
| WATER | 25.0 | 1.1 | 50.00 | 2.775 | Process water, 100 wt% |
| H3PO4 | 20.0 | 1.1 | 40.84 | 1.897 | 20 wt% H3PO4 (8.17 kg/h acid) in water |

### *4.1 Feedstock characterisation*

The oil consists of triglycerides (TG) and diglycerides (DG), with no monoglycerides. On a mole basis the glycerides are 92.5% TG and 7.5% DG. The distribution of acyl groups is 51.6 mol% palmitoyl (P), 36.0 mol% oleoyl (O), 7.3 mol% linoleoyl (LI), 2.7 mol% stearoyl (S) and 2.5 mol% myristoyl (M). This is a palm-oil type composition. The number of ester groups in the fresh oil is 3.74 kmol/h.

The stream table names the glyceride pseudo-components by their acyl groups (for example, PPO for a palmitoyl-palmitoyl-oleoyl triglyceride), and the methyl esters as METHYL-P, METHYL-O and so on.

# 5. SIMULATION RESULTS

***Table 4: Key performance indicators***

| Indicator | Value | Basis |
|---|---|---|
| FAME production | 1,051.8 kg/h (3.72 kmol/h) | ESTCOL FAME side draw |
| FAME ester content | 99.7 wt% | FAME stream composition |
| Glycerol production | 114.2 kg/h, 100 wt% | B1 bottoms |
| Glycerol recovery | 97.7% | Glycerol to GLYCEROL vs. REACOUT |
| One-pass ester conversion | 96.4% | Reactor, ester-group basis (OIL4) |
| Methyl ester yield | 99.7% | Product esters vs. fresh oil ester groups |
| Total methanol to reactor | 247.4 kg/h (7.72 kmol/h) | MIX3 |
| Methanol : oil molar ratio | 6.0 : 1 total; 3.1 : 1 fresh | Fresh oil basis |
| Methanol recovery | 94.0% | MEOHCOL overhead vs. REACOUT |
| Methanol losses | 7.7 kg/h (6.0% of fresh make-up) | WATMEOH, MEOHWAT, FAME |
| NaOH catalyst | 10.0 kg/h (0.95 wt% of oil) | NAOH feed |
| NaOH : H3PO4 molar ratio | 3.0 : 1 | Stoichiometric neutralisation |
| Solid salt removed | 13.7 kg/h Na3PO4 | SOLIDS |
| Net external heating / cooling | +125.0 kW / −13.9 kW | Section 8 |

### *5.1 Reactor performance*

The reactor outlet (REACOUT) contains 3.73 kmol/h of methyl esters. These are formed from 3.73 kmol/h of ester groups converted, which agrees with the ester-group balance to within 0.3%. Methanol is supplied in 2:1 excess on an ester-group basis, and the 3.73 kmol/h consumed leaves 4.00 kmol/h of methanol in the outlet. The reactor outlet contains 116.8 kg/h of glycerol and 10.0 kg/h of unreacted NaOH, which are removed downstream. The 96.4% figure is a one-pass value on the reactor feed, which includes the recycled oil from OILREC2.

The net heat input to the reactor, from the enthalpy balance, is 9.3 kW. This includes the heat of reaction and the heating of the feeds from 32.2 °C and 60.0 °C to the outlet temperature of 60.0 °C.

### *5.2 Methanol recovery and recycle*

MEOHCOL recovers 120.3 kg/h of methanol as a 98.05 wt% overhead, which is 94.0% of the methanol entering the column. The recycle is returned to MIXER2 at 28.8 °C. The remaining 7.7 kg/h of methanol leaves the column in the bottoms (EST1). Of this, 7.6 kg/h is carried into the aqueous stream during washing and ends up in WATMEOH, and the rest leaves with the FAME and MEOHWAT. Methanol consumed in the reaction is 119.4 kg/h.

### *5.3 Water washing*

WASHCOL removes all of the glycerol (116.8 kg/h) and all of the NaOH (10.0 kg/h) from the ester stream into the aqueous stream AQU1. About 98.5% of the methanol entering the washer (7.57 of 7.68 kg/h) also goes to the aqueous stream. The washed ester stream EST4 contains 4.0 kg/h of water, which is removed in ESTCOL. The washing section therefore brings the glycerol and catalyst content of the ester stream to effectively zero.

### *5.4 FAME purification*

ESTCOL separates the washed ester stream (1,098.1 kg/h) into three products: FAME (1,051.8 kg/h), a small overhead vapour MEOHWAT (4.9 kg/h, of which 3.7 kg/h is water) and a bottoms stream OILREC (41.4 kg/h). The methyl ester content of the ester stream in the FAME product is 99.7% of the methyl esters entering ESTCOL. The bottoms contain 0.13 kmol/h of glyceride ester groups, which are recycled to the reactor, so no glyceride is purged from the process.

### *5.5 Catalyst neutralisation and solids removal*

The aqueous stream AQU1 contains 0.250 kmol/h of NaOH. Adding 0.0833 kmol/h of H3PO4 (a 3.0 : 1 molar ratio, the stoichiometric requirement) neutralises all of it, forming 0.0833 kmol/h of Na3PO4 (13.7 kg/h). The FILTER removes this solid completely, so the aqueous stream AQU3 that reaches B1 contains no catalyst. Both the neutralisation and the filtration are modelled as complete.

### *5.6 Glycerol recovery*

B1 receives 247.6 kg/h of aqueous glycerol solution containing 47.2 wt% glycerol. The bottoms (GLYCEROL) are 114.2 kg/h of glycerol at 100 wt% purity, which is 97.7% of the glycerol entering. The overhead (WATMEOH) carries 133.4 kg/h, of which 92.3 wt% is water, 5.7 wt% methanol and 2.0 wt% glycerol. The glycerol lost in the overhead is 2.7 kg/h. The DIRTYH2O side draw carries no flow in the simulation (see Section 9).

## 6. PRODUCT QUALITY

Table 5 compares the simulated FAME product with the EN 14214 limits for the properties the model can predict. The limits are the commonly quoted values for biodiesel and should be confirmed against the current edition of the standard. Properties such as oxidation stability, cold-flow behaviour and cetane number depend on the detailed fatty acid profile and are outside the scope of this simulation.

***Table 5: FAME product quality against EN 14214 limits***

| Property | Simulated value | EN 14214 limit | Assessment |
|---|---|---|---|
| Ester content | 99.7 wt% | ≥ 96.5 wt% | Meets limit |
| Methanol | ≈ 8 mg/kg (0.001 wt%) | ≤ 0.20 wt% | Meets limit |
| Water | ≈ 284 mg/kg | ≤ 500 mg/kg | Meets limit |
| Monoglycerides | 0.27 wt% | ≤ 0.80 wt% | Meets limit |
| Diglycerides | ≈ 0 | ≤ 0.20 wt% | Meets limit |
| Triglycerides | ≈ 0 | ≤ 0.20 wt% | Meets limit |
| Free glycerol | ≈ 0 | ≤ 0.02 wt% | Meets limit |

The FAME product is dominated by methyl palmitate, which is consistent with the palm-type feed. Table 6 gives the methyl ester composition.

***Table 6: Methyl ester composition of the FAME product***

| Methyl ester | Fatty acid | Mass fraction of FAME (wt%) |
|---|---|---|
| Methyl palmitate | C16:0 | 49.3 |
| Methyl oleate | C18:1 | 37.8 |
| Methyl linoleate | C18:2 | 7.6 |
| Methyl stearate | C18:0 | 2.8 |
| Methyl myristate | C14:0 | 2.2 |
| Total methyl esters | | 99.7 |

## 7. MASS BALANCE

Table 7 compares total inlet and outlet mass flows for each unit and for the overall plant. The closure errors are within rounding for all units, which confirms that the stream table is internally consistent.

***Table 7: Mass balance closure by unit***

| Unit | Inlet (kg/h) | Outlet (kg/h) |
|---|---|---|
| MIXER1 (methoxide) | 177.12 | 177.12 |
| PUMP1 | 177.12 | 177.12 |
| MIXER2 | 299.82 | 299.82 |
| B3 (oil mixer) | 1,091.43 | 1,091.43 |
| PUMP2 | 1,091.43 | 1,091.43 |
| HX1 | 1,091.43 | 1,091.43 |
| REACTOR | 1,391.25 | 1,391.25 |

| Unit | Inlet (kg/h) | Outlet (kg/h) |
|---|---|---|
| MEOHCOL | 1,391.25 | 1,391.25 |
| PUMP3 | 1,268.54 | 1,268.54 |
| HX2 | 1,268.54 | 1,268.54 |
| WASHCOL | 1,318.54 | 1,318.54 |
| ESTCOL | 1,098.10 | 1,098.10 |
| B2 | 41.43 | 41.43 |
| NEUTR | 261.28 | 261.28 |
| FILTER | 261.28 | 261.28 |
| B1 | 247.61 | 247.61 |
| Overall plant | 1,317.95 | 1,317.95 |

For the overall plant, the feeds total 1,317.95 kg/h and the products total 1,317.95 kg/h. The products are FAME (1,051.8 kg/h), glycerol (114.2 kg/h), WATMEOH (133.4 kg/h), solids (13.7 kg/h) and MEOHWAT (4.9 kg/h). The DIRTYH2O stream carries no measurable flow.

The sheet also reports a maximum relative error of $3.8 \times 10^{-6}$ across the flowsheet, which indicates the recycle loops have converged.

## 8. ENERGY BALANCE

The energy duties in Table 8 are calculated from the enthalpy flows in the stream table. A positive duty means heat is added to the process stream, and a negative duty means heat is removed. The condenser and reboiler duties of the columns are not reported separately in the data, so each column is shown as a net duty, which is the sum of the reboiler and condenser duties. The reactor duty includes the heat of reaction.

***Table 8: Heat duties by unit***

| Unit | Duty (kW) | Type | Comment |
|---|---|---|---|
| HX1 | +21.9 | Heating | Fresh oil preheat, 25.4 to 60.0 °C |
| REACTOR | +9.3 | Net heat input | Includes heat of reaction |
| ESTCOL | +68.9 | Net heat input | Largest duty; bottoms at 338.7 °C |
| B1 | +24.9 | Net heat input | Glycerol reboil; bottoms at 263.5 °C |
| MEOHCOL | −4.0 | Net heat removal | Vacuum methanol recovery |
| B2 | −9.8 | Cooling | Recycled oil cooled, 338.7 to 25.0 °C |
| HX2 | −0.1 | Trim cooling | Negligible duty, 60.2 to 60.0 °C |
| WASHCOL | 0.0 | Negligible | Water wash, essentially isothermal |

The total net heating is about 125.0 kW and the total net cooling about 13.9 kW. ESTCOL and B1 account for 93.7 kW, or 75% of the total heating, which shows that the two vacuum distillation columns control the utility demand.

The pump hydraulic powers, calculated as volumetric flow multiplied by pressure rise, are 0.017 kW for PUMP1, 0.126 kW for PUMP2 and 0.036 kW for PUMP3, a total of 0.18 kW. Assuming an overall efficiency of 70%, the shaft power is about 0.26 kW. The pumping duty is negligible compared with the heating duty.

## 9. ASSUMPTIONS AND LIMITATIONS

- **Pseudo-components.** Glycerides are represented as lumped pseudo-components named by their acyl groups. The interpretation used in this report is M = myristoyl, P = palmitoyl, S = stearoyl, O = oleoyl and LI = linoleoyl, and METHYL-x denotes the corresponding methyl ester. Triglycerides have three acyl groups, diglycerides two and monoglycerides one.
- **Reaction and kinetics.** The reactor conversion depends on the reaction model used in the simulation. Kinetic parameters, reactor volume and residence time are not included in the data, so the 96.4% conversion has not been validated against experiment or plant data.
- **Operating time.** Annual production figures assume 8,000 operating hours per year, which is a planning assumption and not part of the simulation.

## 10. CONCLUSION

The simulated biodiesel plant converts 1,050 kg/h of a palm-type oil into 1,052 kg/h of FAME with a predicted ester content of 99.7 wt%. The product meets the EN 14214 limits that the model can check, and the overall mass balance closes to within rounding. The reactor achieves a one-pass conversion of 96.4%, and the recycle of methanol and unreacted oil is effective, since 94% of the methanol and 99.7% of the fresh ester groups are recovered.

The main issues to resolve before detailed design are the pressure at the reactor inlet, the methanol and glycerol losses to the aqueous stream, and the thermal conditions in ESTCOL. The most valuable improvement is the heat integration of the recycled oil, which could reduce the heating requirement of HX1 by about 9.8 kW. Taken together, these changes would reduce both utility demand and raw material losses without altering the core reaction and separation scheme.

## APPENDIX A: COMPLETE STREAM SUMMARY

Table A1 gives the temperature, pressure, phase, mass flow, molar flow and average molecular weight of each stream in the flowsheet.

***Table A1: Stream summary***

| Stream | From | To | T (°C) | P (bar) | Phase | Mass (kg/h) | Molar (kmol/h) | MW |
|---|---|---|---|---|---|---|---|---|
| MEOH | — | MIXER1 | 25.0 | 1.00 | Liquid | 127.12 | 3.967 | 32.0 |
| NAOH | — | MIXER1 | 25.0 | 1.00 | Liquid | 50.00 | 2.470 | 20.2 |
| OIL | — | B3 | 25.0 | 1.00 | Liquid | 1,050.00 | 1.277 | 822.0 |
| WATER | — | WASHCOL | 25.0 | 1.10 | Liquid | 50.00 | 2.775 | 18.0 |
| H3PO4 | — | NEUTR | 20.0 | 1.10 | Liquid | 40.84 | 1.897 | 21.5 |
| MIX1 | MIXER1 | PUMP1 | 33.4 | 1.00 | Liquid | 177.12 | 6.438 | 27.5 |
| MIX2 | PUMP1 | MIXER2 | 33.7 | 4.00 | Liquid | 177.12 | 6.438 | 27.5 |
| MIX3 | MIXER2 | REACTOR | 32.2 | 0.20 | Liquid | 299.82 | 10.325 | 29.0 |
| OIL2 | B3 | PUMP2 | 24.9 | 0.20 | Liquid | 1,091.43 | 1.359 | 802.9 |
| OIL3 | PUMP2 | HX1 | 25.4 | 4.00 | Liquid | 1,091.43 | 1.359 | 802.9 |

| Stream | From | To | T (°C) | P (bar) | Phase | Mass (kg/h) | Molar (kmol/h) | MW |
|---|---|---|---|---|---|---|---|---|
| OIL4 | HX1 | REACTOR | 60.0 | 4.00 | Liquid | 1,091.43 | 1.359 | 802.9 |
| REACOUT | REACTOR | MEOHCOL | 60.0 | 4.00 | Liquid | 1,391.25 | 11.685 | 119.1 |
| MEOHREC | MEOHCOL | MIXER2 | 28.8 | 0.20 | Liquid | 122.71 | 3.888 | 31.6 |
| EST1 | MEOHCOL | PUMP3 | 60.0 | 0.20 | Liquid | 1,268.54 | 7.797 | 162.7 |
| EST2 | PUMP3 | HX2 | 60.2 | 1.10 | Liquid | 1,268.54 | 7.797 | 162.7 |
| EST3 | HX2 | WASHCOL | 60.0 | 1.10 | Liquid | 1,268.54 | 7.797 | 162.7 |
| EST4 | WASHCOL | ESTCOL | 57.8 | 1.00 | Liquid | 1,098.10 | 4.042 | 271.7 |
| FAME | ESTCOL | — | 148.3 | 0.10 | Liquid | 1,051.79 | 3.749 | 280.6 |
| MEOHWAT | ESTCOL | — | 148.3 | 0.10 | Vapor | 4.88 | 0.211 | 23.1 |
| OILREC | ESTCOL | B2 | 338.7 | 0.20 | Liquid | 41.43 | 0.082 | 505.5 |
| OILREC2 | B2 | B3 | 25.0 | 0.20 | Liquid | 41.43 | 0.082 | 505.5 |
| AQU1 | WASHCOL | NEUTR | 60.9 | 1.00 | Liquid | 220.45 | 6.530 | 33.8 |
| AQU2 | NEUTR | FILTER | 50.0 | 1.10 | Liquid | 261.28 | 8.427 | 31.0 |
| AQU3 | FILTER | B1 | 50.0 | 1.10 | Liquid | 247.61 | 8.344 | 29.7 |
| SOLIDS | FILTER | — | 50.0 | 1.10 | Liquid | 13.68 | 0.083 | 164.0 |
| WATMEOH | B1 | — | 70.9 | 0.40 | Liquid | 133.42 | 7.104 | 18.8 |
| GLYCEROL | B1 | — | 263.5 | 0.50 | Liquid | 114.18 | 1.240 | 92.1 |
| DIRTYH2O | B1 | — | 70.9 | 0.40 | Liquid | 0.00 | 0.000 | 274.5 |